

Messy shop exports in, a tested board out every morning

An ELT pipeline for Copper Kettle, an invented chain of three coffee shops, rebuilt daily by GitHub Actions.

The problem

Sales sit in two POS export formats with duplicates and refunds, online orders arrive as webhooks with retries and cancellations, and nobody trusts the weekly spreadsheet built by hand from them.

What I built

- Real daily weather from the Open-Meteo API, with an archive fallback for older days
- Raw files loaded untouched into DuckDB
- dbt staging models into dimensions, facts and a daily board that ranks the shops
- Uniqueness, not-null, relationship, reconciliation and freshness tests
- Daily Actions run that publishes the KPI page, docs and lineage

Measured results

53/53

dbt nodes pass, including 41 data tests, before the board publishes

4/4

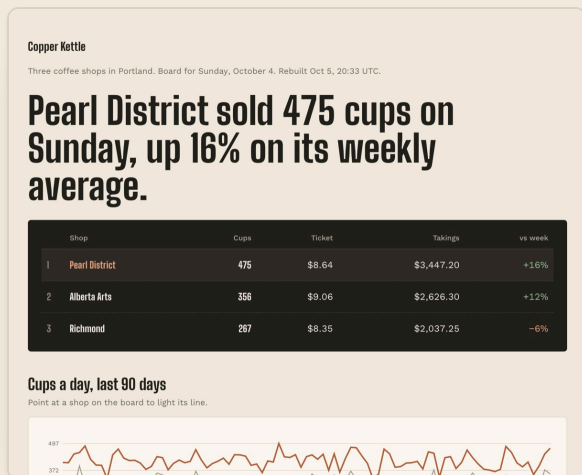
source freshness checks pass on every run

108,790

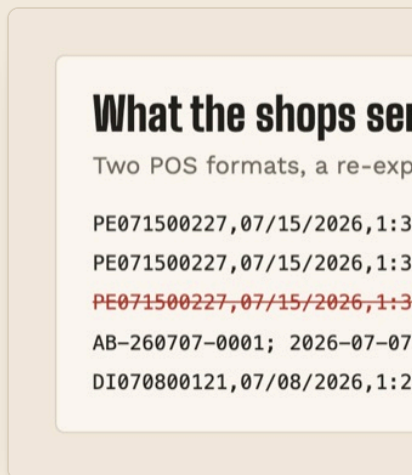
raw records cleaned into one board each morning

3 issues

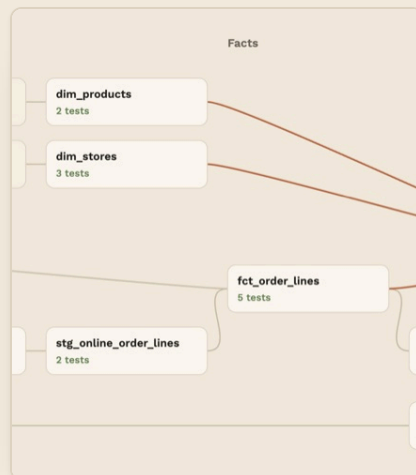
caught by tests while building, each fixed and now tested



The daily board, live on GitHub Pages



What the shops send



Model lineage

Stack

Python

DuckDB

dbt

GitHub Actions

Open-Meteo API

Live vanthienha199.github.io/data-pipeline-demo/
Code github.com/vanthienha199/data-pipeline-demo